

# MyHealth Care

## Complete Project & Feature Report — What Was Built

|            |                                                                   |        |                                          |
|------------|-------------------------------------------------------------------|--------|------------------------------------------|
| PROJECT    | MyHealth Care — Clinical Records, AI Summaries & Smart Scheduling | STATUS | Core build complete + 3 follow-up phases |
| TEAM       | Ali Mohamed Jaafar Mohamed · Mohammed A. Redha Meftah             | DATE   | 24 September 2026                        |
| SUPERVISOR | Dr. Amal Ghanim                                                   | TESTS  | 178 automated tests passing              |

### About This Document

This report brings together everything built for MyHealth Care: an offline-first clinical records app with an AI health summary, a machine-learning no-show predictor that ranks booking slots, and full staff/admin workflows — built for a University of Bahrain senior project.

It is organised by build phase, from project setup through the three post-review phases (polish, full redesign, patient services & documents), followed by open items and supporting documents.

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## PART 0 • SETUP

## Project Setup

Foundations: framework, local database, project hygiene.

### Flutter

- [x] Create Flutter project (Windows, Android, Web, iOS)
- [x] App name “MyHealth Care” + app icons (all platforms)
- [x] All packages (Riverpod, Drift, go\_router, dio, fl\_chart, ...)
- [x] Feature-first folder structure (domain / data / services)
- [x] Theme — colours, clinical status colours, fonts, spacing
- [x] Routing — go\_router, one shell per role
- [x] ProviderScope + dependency registry

### Local Database (no Firebase, offline-first)

- [x] Drift + SQLite on Windows
- [x] Drift + SQLite on Web (WASM + OPFS)
- [ ] Drift + SQLite on Android — blocked on hypervisor driver
- [x] Result / Failure types, no exceptions across layers

### Project Hygiene

- [x] Strict lint rules, flutter analyze clean
- [x] Git repository + .gitignore (no API keys committed)
- [x] README (run, seed, add AI key)

## PART 1 · DATABASE

## Database

20+ tables, a full domain layer, foreign-key-enforced relations and two years of synthetic clinical data.

### Tables

- [x] Users, Patient Profiles, Staff Profiles, Departments
- [x] Appointments, Schedule Templates, Reminders
- [x] Medical Records, Lab Values, Vitals, Medications
- [x] AI Summaries, Staff Tasks, Risk Flags
- [x] Audit Log, App Settings
- [x] Invoices, Payment Methods, Notifications, Feedback, AI Usage Log
- [x] Sick-Leave Certificates, Care Messages, Home-Visit Requests

### Domain Layer

- [x] Entities (User, Patient, Staff, Appointment, Record, ...)
- [x] Repository interfaces (swap point for a future server)
- [x] Repository implementations
- [x] Registered in the dependency registry

### Relations (foreign keys)

- [x] Patient -> Appointments, Records, Vitals, Medications
- [x] Appointment -> Patient, Staff, Department, Payment, Review
- [x] Staff -> Appointments, Schedule Templates, Tasks
- [x] Record -> Lab Values
- [x] Risk Flag -> Patient · Staff Task -> Staff + Patient
- [x] Department -> Staff

### Synthetic Data (no real patients)

- [x] Clinical vocabulary (conditions, meds, lab ranges)
- [x] Seeder: 5 departments, 12 doctors, 60 patients, ~2 yrs history
- [x] Idempotent re-seed + reset demo data
- [x] Database ERD exported to docs/erd.png

## PART 2 · CORE

## Authentication & Patient Core

Secure login and the full patient experience — timeline, vitals, medications, profile.

### Authentication

- [x] Salted password hashing (PBKDF2-HMAC-SHA256)
- [x] Login screen
- [x] Patient self-registration
- [x] Session persistence (stay logged in after restart)
- [x] Role-gated routing (patient / staff / admin)
- [ ] Logout + switch-user UI polish

### Patient Screens

- [x] Home (next appointment, active meds, AI summary card)
- [x] Health timeline (records + vitals, grouped by month)
- [x] Timeline filters + search
- [x] Record detail (lab values flagged vs reference range)
- [x] Vitals list + manual entry
- [x] Vitals charts (BP, weight, glucose, heart rate)
- [x] Medications list (current / past)
- [x] Profile & settings
- [ ] PDF report import (pick, extract, save as record)
- [ ] Attachment viewer

Uses local password auth, not phone/OTP — real SMS verification would be added for production.

## PART 3 · AI

## AI Health Summary

Research Question 1 — a narrative summary generated from each patient's real data, with a mock fallback so the app never depends on a live key.

### Engine

- [x] AiService interface + response models
- [x] Patient context builder (token budget)
- [x] Mock AI service — deterministic, from real patient data
- [x] Summary prompt + strict JSON contract + tolerant parser
- [x] Live AI service — Google Gemini (free tier), retry, timeout
- [x] Fallback: live AI fails -> drop to mock automatically
- [x] API key in secure storage, never committed

### Experience

- [x] Provider picks mock vs live from settings
- [x] Summary cache keyed on input hash
- [x] AI summary screen (narrative + events + trends + flags)
- [x] Summary card on home screen + regenerate
- [x] Key events shown as markers on the timeline
- [x] Trends link to the matching vitals chart
- [x] “Not medical advice” safety banner on every AI screen
- [x] Model id + prompt version + timestamp stored per summary

## PART 4 · ML

## No-Show Model & Smart Scheduling

Research Question 2 — a logistic regression model trained offline in Python and run on-device to rank booking slots by predicted no-show risk.

### ML Machine Learning (Python, trained offline)

Trained on a synthetic dataset with 11 engineered features.

|         |                                                                          |
|---------|--------------------------------------------------------------------------|
| Dataset | ~8,500 synthetic appointments                                            |
| Model   | Class-balanced logistic regression                                       |
| Result  | ROC-AUC 0.75, recall 0.62 — beats majority-class and heuristic baselines |
| Export  | Model weights exported to JSON, bundled as an app asset                  |

### On-Device Inference (Dart)

- [x] Feature extractor — identical encoding to Python
- [x] Predictor — scale, sigmoid, risk band
- [x] Per-feature contribution breakdown (explainability)
- [x] Python <-> Dart parity test within 1e-6

### Reminders

- [x] Notification abstraction + in-app fallback (works on web)
- [x] Risk-adaptive reminder escalation (1/2/3 by risk band)

### Booking

- [x] Open-slot generator (templates minus booked)
- [x] Booking wizard (dept -> doctor -> date -> slot -> confirm)
- [x] Slots ranked by predicted risk, plain-language reason
- [x] My appointments — cancel, reschedule
- [x] Predicted risk + band stored on the appointment
- [ ] AI-written slot rationale — skipped, model reasons suffice

## PART 5 · WORKFLOWS

## Staff & Admin

Research Question 3 — a rule engine that works with AI switched off, plus full staff clinical tools and admin platform management.

### Rule Engine (AI-independent)

- [x] Risk detection — abnormal vitals, critical labs, med gaps, overdue follow-ups
- [x] Severity scoring (info / warning / urgent) + de-duplication
- [x] Rule-based task generator

### Staff App

- [x] Staff shell (Dashboard / Patients / Tasks / Schedule / Profile)
- [x] Dashboard — schedule, risk flags, task preview, scan panel
- [x] Patient search + list, patient chart
- [x] Add clinical note, prescribe medication
- [x] Enter lab result (auto-classified vs reference range)
- [x] AI task prioritisation blended with rule score
- [x] Task board with per-task rationale, week schedule grid
- [x] Panel analytics — no-show rate, cancellations, utilisation

### Admin Dashboard

- [x] User management — create staff, activate/deactivate, reset password
- [x] Departments — create / rename / describe
- [ ] Schedule-template editor — deferred, templates are seeded
- [x] AI settings — API key, model, mock toggle, blend weight, re-seed
- [x] System analytics + audit log viewer

## PART 6 · QUALITY

## Testing & Results

Every number in the report is backed by a test or a measurement.

### Automated Tests

- [x] Unit — predictor, features, risk rules, password hashing
- [x] Unit — repositories on an in-memory database
- [x] Unit — AI parsing (valid, malformed, empty, timeout)
- [x] Widget — login, timeline, booking, staff dashboard
- [x] Integration — seed -> patient books -> staff sees it

### Cross-Platform & Measurements

- [x] Windows + Web release builds green
- [ ] Android smoke test — blocked on hypervisor driver
- [ ] Usability study (5–8 people, SUS) — protocol ready, needs participants
- [x] Performance — timeline load, AI latency, cold start, DB size
- [x] Accessibility — contrast, tap targets, labels, text scaling

**Result: 178 tests, all passing. flutter analyze clean.**

## PART 8 · POLISH

## Post-Review Polish

Gaps and bugs found while walking the build with the supervisor.

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- [x] Sign out available across the whole admin app
- [x] User management — Add follows selected tab + search bar
- [x] Departments — delete (refused while still referenced)
- [x] Admin dashboard landing screen (counts, health, quick links, audit)
- [x] Fix: staff Schedule tab crashed on an empty week
- [x] Staff profile panel + theme (light/dark) + language (EN/AR)
- [x] Patient profile as a 5th bottom-nav tab (both roles)
- [x] Booking flow shows every slot open-to-close; weekend guard
- [x] Clinic day extended to 08:00–20:00
- [x] Appointments tab reorganised
- [x] New “MyHealth Care” logo on every platform
- [x] Admin profile screen (account, audit, analytics, forecast, AI settings, AI log, preferences, clinic hours)

## PART 9 • REDESIGN

## Full “v2” Redesign

One product, three experiences (patient / staff / admin) reworked together.

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- [x] Design system: indigo-violet palette, flat surfaces + hairline borders, motion tokens, size-specific type
- [x] Shared components: AppCard, SectionHeader, MetricTile, InlineBanner, empty/loading/error states
- [x] ~24 screens rebuilt across all three roles
- [x] DESIGN.md rewritten to v2 (identity, palette, IA, component catalogue)
- [ ] R4 IA reshuffle + 3 remaining theme-only screens — deferred

## PART 10 · SERVICES

## Patient Services & Documents

A patient can get every routine clinic document and service without a phone call.

### Documents (offline PDF)

- [x] PDF engine — letterhead, identity strip + allergy banner, footer
- [x] Vital-signs report (PDF) — latest snapshot + full history
- [x] Radiology results — imaging records, list + one-page report PDF
- [x] Sick-leave certificate (PDF) — doctor-issued, view + download

### Services

- [x] Visited doctors — care team from appointment history, “book again”
- [x] Allergies — read-only, high-visibility view + alert strips
- [x] Ask your doctor — patient <-> doctor message threads + staff inbox
- [x] Home health care — visit requests + admin triage queue

### Data

- [x] 3 new tables (schema 11, seedVersion 13)
- [x] Seed data + repository tests + end-to-end flow tests

## PART 7 · IN PROGRESS

## Report & Presentation

The last phase before defense — not yet complete.

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- [ ] Architecture, use-case, sequence diagrams
- [ ] Screenshots — all screens, both roles, light + dark
- [ ] Results chapter — ML metrics, baselines, SUS, performance
- [ ] Demo script, rehearsed offline with the mock AI
- [ ] Presentation deck + rehearsal

## APPENDIX

## Every Screen in the App

All routes defined in the app router (`lib/app/router.dart`), grouped by area. This is the literal screen inventory behind the feature list above.

### Auth (public, before sign-in)

| Screen          | Route            |
|-----------------|------------------|
| Splash          | /                |
| Login           | /login           |
| Register        | /register        |
| Forgot Password | /forgot-password |
| Reset Password  | /reset-password  |

### Patient — bottom nav: Home / Nutrition / Appointments / Records / Profile

| Screen                           | Route                           |
|----------------------------------|---------------------------------|
| Home                             | /patient/home                   |
| Vitals (list + entry + charts)   | /patient/home/vitals            |
| Billing                          | /patient/home/billing           |
| Notifications                    | /patient/home/notifications     |
| Messages (Ask Your Doctor)       | /patient/home/messages          |
| Message thread                   | /patient/home/messages/:staffId |
| Home Health Visit request        | /patient/home/home-visit        |
| AI Health Summary                | /patient/summary                |
| Booking wizard                   | /patient/appointments/book      |
| Nutrition calculator + meal plan | /patient/nutrition              |
| Appointments (My Appointments)   | /patient/appointments           |
| Visited Doctors / care team      | /patient/appointments/doctors   |
| Health Records / Timeline        | /patient/timeline               |
| Radiology / imaging results      | /patient/timeline/imaging       |
| Allergies                        | /patient/timeline/allergies     |
| Sick-leave certificate           | /patient/timeline/sick-leave    |
| Record detail                    | /patient/timeline/record/:id    |
| Profile hub                      | /patient/settings               |
| Personal info                    | /patient/settings/personal      |
| Health details                   | /patient/settings/health        |
| Wallet                           | /patient/settings/wallet        |
| Preferences (theme / language)   | /patient/settings/preferences   |
| Family network                   | /patient/settings/family        |
| Care Navigator (AI chat overlay) | overlay on every patient screen |

## APPENDIX

## Every Screen in the App (continued)

### Staff — bottom nav: Dashboard / Patients / Tasks / Schedule / Profile

| Screen                           | Route                              |
|----------------------------------|------------------------------------|
| Staff Dashboard                  | /staff/dashboard                   |
| Notifications                    | /staff/dashboard/notifications     |
| Inbox (Ask Your Doctor)          | /staff/dashboard/inbox             |
| Message thread                   | /staff/dashboard/inbox/:patientId  |
| Clinical Scribe (AI note-taking) | /staff/scribe                      |
| Consultation (per appointment)   | /staff/consultation/:appointmentId |
| Patients (search + list)         | /staff/patients                    |
| Patient chart                    | /staff/patients/:id                |
| Patient AI summary               | /staff/patients/:id/summary        |
| Task board                       | /staff/tasks                       |
| Week schedule                    | /staff/schedule                    |
| Profile hub                      | /staff/profile                     |
| Account                          | /staff/profile/account             |
| My activity                      | /staff/profile/activity            |
| Staff directory                  | /staff/profile/directory           |
| Panel analytics                  | /staff/profile/analytics           |
| Preferences (theme / language)   | /staff/profile/preferences         |

## APPENDIX

## Every Screen in the App (continued)

### Admin — bottom nav: Dashboard / Users / Departments / Billing / Profile

| Screen                         | Route                              |
|--------------------------------|------------------------------------|
| Admin Dashboard                | /admin/dashboard                   |
| Notifications                  | /admin/dashboard/notifications     |
| Home-visit triage queue        | /admin/dashboard/home-visits       |
| Referral requests              | /admin/dashboard/referral-requests |
| Appointments (operational)     | /admin/appointments                |
| Feedback                       | /admin/feedback                    |
| User management                | /admin/users                       |
| Departments                    | /admin/departments                 |
| Billing                        | /admin/billing                     |
| Profile hub                    | /admin/profile                     |
| Account                        | /admin/profile/account             |
| Audit log                      | /admin/profile/audit               |
| System analytics               | /admin/profile/analytics           |
| Forecast                       | /admin/profile/forecast            |
| AI settings                    | /admin/profile/ai                  |
| AI usage log                   | /admin/profile/ai-log              |
| Preferences (theme / language) | /admin/profile/preferences         |
| Clinic hours                   | /admin/profile/clinic-hours        |

## OPEN ITEMS

## What's Left

Everything still open across the project, and what closes it.

| Item                              | Status   | To close                                                          |
|-----------------------------------|----------|-------------------------------------------------------------------|
| Android build                     | Blocked  | Run the AEHD hypervisor driver installer, or use a physical phone |
| PDF report import                 | Deferred | Wire file picker -> text extractor -> new record                  |
| Schedule-template editor          | Deferred | Add a schedule-template repository                                |
| R4 IA reshuffle + 3 theme screens | Deferred | Finish the v2 redesign                                            |
| Usability study results           | Pending  | Run the protocol in docs/usability_study.md                       |
| Live Gemini output                | Dormant  | Add a free API key in Admin -> AI Settings, turn Mock off         |
| Message / triage notifications    | Deferred | Notify on doctor reply or home-visit decision                     |

### Supporting Documents (docs/)

- MyHealthCare\_Feature\_Report.pdf — full feature & implementation report
- test\_accounts.md — every demo login (1 admin, 12 staff, 60 patients, password: password)
- ai\_setup.md — how to enable the free Gemini API
- platform\_smoke.md — per-platform build status
- performance.md — performance measurements
- usability\_study.md — SUS study protocol
- ml\_results.md — no-show model metrics
- erd.png — database diagram